MAHATMA GANDHI UNIVERSITY, KOTTAYAM
BSc (Honours) Food Technology and Quality Assurance
Third Semester Examination (2024 Admission onwards)
Course Code- MG3DSCFTQ201-Food Analytical Instrumentation

Duration:**1.5Hours** Maximum Marks: **50**

Part **A**
Short Answer Type Questions
Answer any **5** questions
Each question carries **2** Marks

1.	Define chromatography.	[U]	[CO1]
2.	Degasser in HPLC	[A]	[CO2]
3.	Cite stationary phase in GC	[K]	[CO2]
4.	List out applications of spectrophotometer.	[K]	[CO2]
5.	Centrifugation	[A]	[CO3]
6.	Electrophoretic mobility.	[U]	[CO5]
7.	Buffers used in electrophoresis.	[U]	[CO5]

(5x2=10 Marks)

Part **B**
Short Essay Type Questions
Answer any **5**questions
Each question carries **4**Marks

8.	Explain Beer's law.	[U]	[CO1]
9.	Add a short note on monochromator.	[K]	[CO2]
10.	Differentiate pre- column and analytical column in HPLC.	[A]	[CO2]
11.	What is the separation principle in size exclusion chromatography?	[U]	[CO3]
12.	Explain the types of atomizers in AAS.	[U]	[CO5]
13.	What is the principle behind electrophoresis?	[U]	[CO5]
14.	Add a note on types of rotor heads used in centrifugation.	[U]	[CO5]

(5x4=20 Marks)

Part **C**
Long Essay Type Questions
Answer any **2** question

Each question carries **10** Marks

15.	Explain working of single beam UV-VIS absorption spectroscopy.	[U]	[CO1]
16.	Discuss in detail about TLC with neat diagram.	[U]	[CO2]
17.	Discuss about different modes of centrifugation.	[U]	[CO4]
18.	Explain about SDS-PAGE. Mention its applications.	[U]	[CO5]

(2x10=20 Marks)